NATIONALS PLAYER

STATS ANALYSIS

2019 Season · Excel & Access Database Project

Ben Gibson

PROJECT OVERVIEW

Dataset

50 Washington Nationals players — 2019 season batting and pitching stats

Tools Used

- Microsoft Excel for calculations and visualization
- Microsoft Access for database design and queries

Key Metrics Analyzed

- Batting AVG, Home Runs (HR), RBIs
- ERA and Innings Pitched
- Comparison to league average (.275 AVG)

PlayerID	FirstName	LastName	H	HR	AB	AVG	OBP	IP	ER	ERA	RBI	Diff from League AVG	Above/Below Avg	All-Star?
1	Austin	Adams	0	0	0			1.0	1	9.00	0		Above Average	
2	Matt	Adams	70	20	310	0.226	0.276	0.0	0		56	0.226	Below Average	
3	Kyle	Barraclough	0	0	0			25.2	19	6.79	0		Above Average	
4	Tres	Barrera	0	0	2	0.000	0.000	0.0	0		0	0.000	Below Average	
5	Aaron	Barrett	0	0	0			2.1	4	17.14	0		Above Average	
6	Michael	Blazek	0	0	0			5.0	4	7.20	0		Above Average	
7	James	Bourque	0	0	0			0.2	4	180.00	0		Above Average	
8	Asdrúbal	Cabrera	40	6	124	0.323	0.404	0.0	0		40	0.323	Above Average	
9	Patrick	Corbin	6	0	65	0.092	0.132	202.0	73	3.25	4	0.092	Below Average	
10	Wilmer	Difo	33	2	131	0.252	0.315	0.0	0		8	0.252	Below Average	
11	Sean	Doolittle	0	0	3	0.000	0.000	60.0	27	4.05	0	0.000	Below Average	
12	Brian	Dozier	99	20	416	0.238	0.340	1.0	2	18.00	50	0.238	Below Average	
13	Adam	Eaton	158	15	566	0.279	0.365	0.0	0		49	0.279	Above Average	
14	Roenis	Elías	0	0	1	0.000	0.000	3.0	3	9.00	0	0.000	Below Average	
15	Erick	Fedde	2	0	15	0.133	0.133	78.0	39	4.50	0	0.133	Below Average	
16	Yan	Gomes	70	12	314	0.223	0.316	0.0	0		43	0.223	Below Average	
17	Matt	Grace	0	0	3	0.000	0.000	46.2	33	6.43	0	0.000	Below Average	
18	Javy	Guerra	0	0	3	0.000	0.000	53.2	29	4.91	0	0.000	Below Average	
19	Jeremy	Hellickson	1	0	9	0.111	0.273	39.0	27	6.23	0	0.111	Below Average	
20	Daniel	Hudson	0	0	0			25.0	4	1.44	0		Above Average	
21	Dan	Jennings	0	0	0			4.2	7	15.00	0		Above Average	
22	Howie	Kendrick	115	17	334	0.344	0.395	0.0	0		62	0.344	Above Average	
23	Carter	Kieboom	5	2	39	0.128	0.209	0.0	0		2	0.128	Below Average	
24	Kyle	McGowin	1	0	3	0.333	0.333	16.0	18	10.13	0	0.333	Above Average	
25	Justin	Miller	0	0	0			15.2	7	4.14	0		Above Average	
26	Jake	Noll	2	0	12	0.167	0.231	0.0	0		2	0.167	Below Average	
27	Gerardo	Parra	47	8	188	0.250	0.300	0.0	5		42	0.250	Below Average	
28	Tanner	Rainey	0	0	1	0.000	0.000	48.1	21	3.93	0	0.000	Below Average	
29	Raudy	Read	1	0	11	0.091	0.091	0.0	15		0	0.091	Below Average	
30	Anthony	Rendon	174	34	545	0.319	0.412	0.0	0		126	0.319	Above Average	
31	Víctor	Robles	139	17	546	0.255	0.326	0.0	0		65	0.255	Below Average	
32	Fernando	Rodney	0	0	0			33.1	16	4.35	0		Above Average	
33	Trevor	Rosenthal	0	0	0			6.1	39	57.54	0		Above Average	
34	Joe	Ross	2	0	19	0.105	0.150	64.0	0	0.00	0	0.105	Below Average	
35	Aníbal	Sánchez	7	0	31	0.226	0.250	166.0	71	3.85	1	0.226	Below Average	
36	Adrián	Sanchez	6	0	52	0.115	0.115	0.0	0		1	0.115	Below Average	
37	Max	Scherzer	10	0	55	0.182	0.182	172.1	56	2.93	2	0.182	Below Average	
38	Tony	Sipp	0	0	0			21.0	11	4.71	0		Above Average	
39	Juan	Soto	153	34	542	0.282	0.401	0.0	0		110	0.282	Above Average	
40	Andrew	Stevenson	11	0	30	0.367	0.486	0.0	0		0	0.367	Above Average	
41	Stephen	Strasburg	12	1	72	0.167	0.200	209.0	77	3.32	10	0.167	Below Average	
42	Hunter	Strickland	0	0	0			21.0	12	5.14	0		Above Average	
43	Wander	Suero	0	0	1	0.000	0.000	71.1	36	4.56	0	0.000	Below Average	
44	Kurt	Suzuki	74	17	280	0.264	0.324	0.0	0		63	0.264	Below Average	
45	Michael A.	Taylor	22	1	88	0.250	0.305	0.0	0		3	0.250	Below Average	
46	Trea	Turner	155	19	521	0.298	0.353	0.0	0		57	0.298	Above Average	

EXCEL ANALYSIS

Calculations

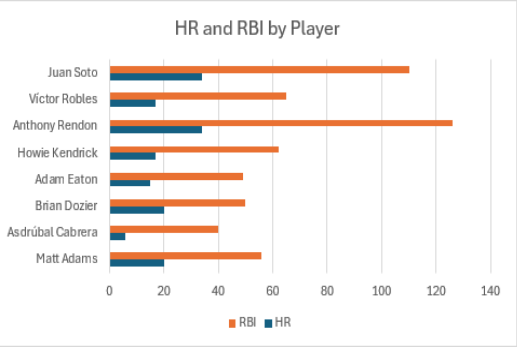
- Batting AVG, ERA, and difference from league average for each player
- Conditional formatting: green = above average, red = below average

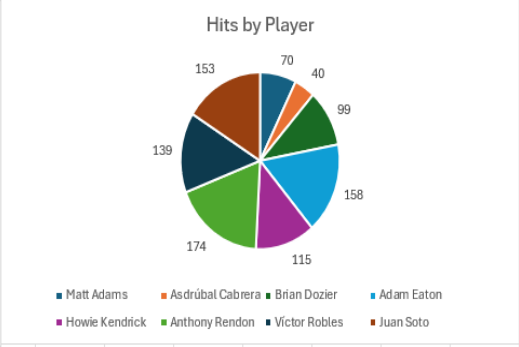
Functions Used

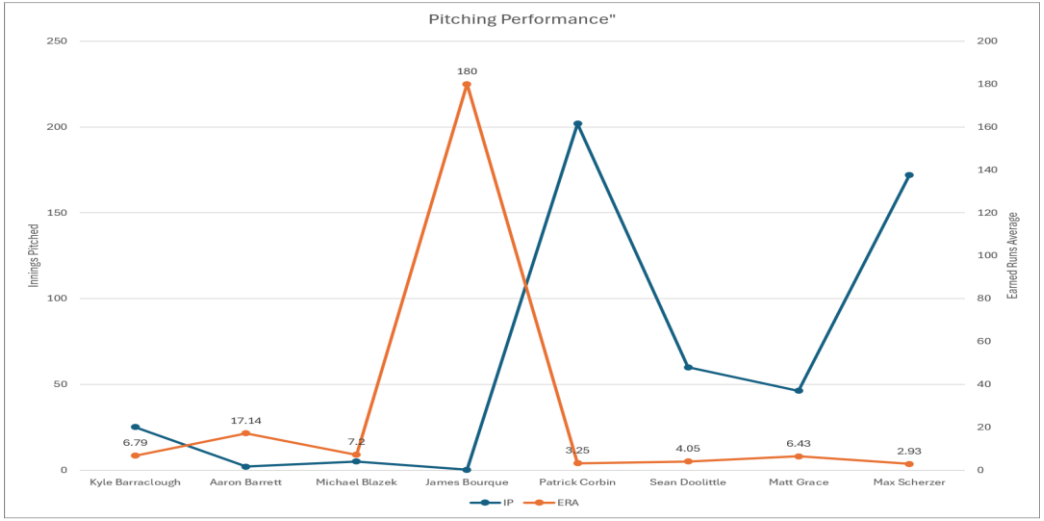
- VLOOKUP, IF, Absolute References, Data Validation

Charts Built

- Bar chart: HR & RBI by player
- Pie chart: Batting distribution
- Line chart: Pitchers' ERA by player







ACCESS DATABASE — TOP PLAYERS ANALYSIS

Database Structure

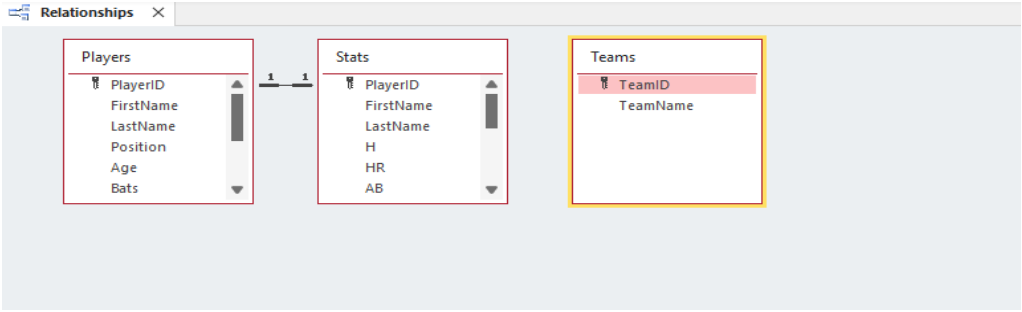
- 3 tables: Players, Stats, Teams
- Relationship: PlayerID → PlayerID (Players ↔ Stats)

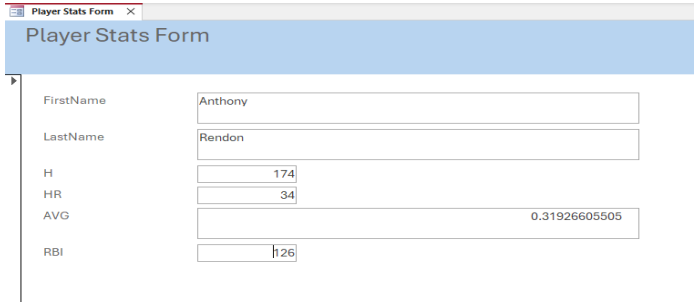
Queries

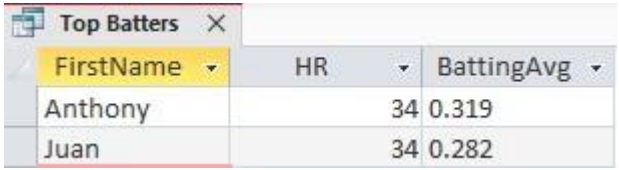
- Filtered players with >20 HRs and AVG above .275 (league average)

Forms & Reports

- Built input forms and summary reports to organize and display player data







Top Batters

FirstName	HR	BattingAvg
Anthony	34	0.319
Juan	34	0.282

SUMMARY & KEY FINDINGS

Key Findings

- Juan Soto and Anthony Rendon led the team with 34 HRs each, both exceeding the .275 league batting average
- Pitching data revealed Max Scherzer as the top performer with the lowest ERA among starters

Skills Demonstrated

- Excel: VLOOKUP, IF, conditional formatting, data validation, charting
- Access: table design, relational keys, query building, forms, reports
- Integrated both tools to move from raw data to structured insights

A	B	C
PLayerID	LastName via VLOOKUP	AVG
1	Adams	
2	Adams	0.226
3	Barraclough	
4	Barrera	0.000
5	Barrett	
6	Blazek	
7	Bourque	
8	Cabrera	0.323
9	Corbin	0.092
10	Difo	0.252
11	Doolittle	0.000
12	Dozier	0.238
13	Eaton	0.279
14	Elías	0.000
15	Fedde	0.133
16	Gomes	0.223
17	Grace	0.000
18	Guerra	0.000
19	Hellickson	0.111
20	Hudson	
21	Jennings	
22	Kendrick	0.344
23	Kieboom	0.128
24	McGowin	0.333
25	Miller	
26	Noll	0.167
27	Parra	0.250
28	Rainey	0.000
29	Read	0.091
30	Rendon	0.319
31	Robles	0.255
32	Rodney	
33	Rosenthal	
34	Ross	0.105
35	Sánchez	0.226
36	Sanchez	0.115
37	Scherzer	0.182
38	Sipp	
39	Soto	0.282
40	Stevenson	0.367
41	Strasburg	0.167
42	Strickland	
43	Suero	0.000
44	Suzuki	0.264
45	Taylor	0.250
46	Turner	0.298
47	Venters	
48	Voth	0.111
49	Williams	
50	Zimmerman	0.257